

Perform a Technical Review

This training resource describes how to perform a Technical Review for HPL in the Proposal System

Requirements:

Able to login to the Proposals System, <https://proposal.facilities.rl.ac.uk>

Have been assigned a Technical Reviewer



Science and
Technology
Facilities Council

Contents:

- **Login**
 - Select your role
 - View My Proposals
- **Add to Technical Review**
 - [View Proposal information](#)
 - [Add your comments to Technical Review](#)
 - [Save your comments](#)



Technical Reviews are likely to be undertaken by several people as there are numerous sections where contributions are needed.

FAP Reviews will be completed by HPL FAP members.



Science and
Technology
Facilities Council

Login to your Facilities account – <https://proposal.facilities.rl.ac.uk/>

Enter your Email address/Fed ID and Password
Click Sign In



Please Login

Email / Fed ID

mei.ling@stfc.ac.uk

Password

[Create an account](#)

[Forgot password?](#)



Success!



Sign In

Go to Proposals –

UKRI

Science and Technology Facilities Council

Dashboard

Reviewer

All proposals

Call

All

Proposals

☐	Actions	Proposal ID	Title
☐		26110021	Real-Time Observation of Ca..
☐		24210015	X-ray Emission from Laser-Ir..
☐		24220011	Ultrafast Laser-Induced Phas..
☐		26110018	Time-Resolved Spectroscop...
☐		26110009	Laser Driven Ion Acceleration..

Assigned Roles

User roles

Action	Role	Description
IN USE	User	Submit and view your proposals
USE	User Officer	Manage the system
USE	FAP Secretary	View and manage your FAP panel and submit FAP reviews
USE	FAP Reviewer	Submit and view your FAP reviews
USE	Experiment Scientist	Edit and view your technical feasibility reviews and view proposals for your experimental area

CLOSE

Mei Ling
mei.ling@stfc.ac.uk

Manage account

Roles

Logout

User

Technical Status

REVIEW

5 days

Reed,Jim

REVIEW

5 days

Reed,Jim

REVIEW

5 days

Ling,Mei

REVIEW

3 days

Ling,Mei

You will need to have the role of **Experimental Scientist**.
You can get to this via the people icon.

Go to Proposals –







Reviewer		Call		Experimental Area			Status			
All proposals		All		All			All			
Proposals						Internal Reviewer				
						Search				
	Actions	Proposal ID	Title	Principle Investigator	IN USE	Submitted	Status	Technical Status	Technical time allocation	Assigned technical reviewer
		26110021	Real-Time Observation of Ca..	O'Connor,Liam	liamoconnor@utoronto.ca	Yes	FAP_AND_FEASIBILITY_REVIEW		3 days	Ling,Mei
		24210015	X-ray Emission from Laser-Ir..	Torres,Camila	camila.torres@imperial.ac.uk	Yes	Submitted (locked)		3 days	Reed,Jim
		24220011	Ultrafast Laser-Induced Phas..	Silva,Ana	ana.silva@warwick.ac.uk	Yes	Submitted (locked)		5 days	Reed,Jim
		26110018	Time-Resolved Spectroscop...	Rossi,Luca	luca.rossi@uct.ac.za	Yes	FAP_AND_FEASIBILITY_REVIEW		5 days	Ling,Mei
		26130009	Laser Driven Ion Acceleration..	O'Connor,Liam	liamoconnor@utoronto.ca	Yes	FAP_AND_FEASIBILITY_REVIEW		3 days	Ling,Mei

All submitted proposals and all proposals on the call will appear here

Add information to a Technical Review

UKRI

Science and Technology Facilities Council

Dashboard

Reviewer

My proposals

My proposals

All proposals

Call

All

Experimental Area

All

Status

All

more

Select My proposals to show the proposals you need to review.

☐	Actions	Proposal ID	Title	Principle Investigator	PI Email	Submitted	Status	Technical Status	Technical time allocation	Assigned technical reviewer
☐		26110021	Real-Time Observation of Ca..	O'Connor,Liam	liamoconnor@utoronto.ca	Yes	FAP_AND_FEASIBILITY_REVIEW		3 days	Ling,Mei
☐		26110018	Time-Resolved Spectroscop...	Rossi,Luca	luca.rossi@uct.ac.za	Yes	FAP_AND_FEASIBILITY_REVIEW		5 days	Ling,Mei
☐		26110009	Laser Driven Ion Acceleration..	O'Connor,Liam	liamoconnor@utoronto.ca	Yes	FAP_AND_FEASIBILITY_REVIEW		3 days	Ling,Mei

Proposals to be reviewed are designated by an Edit icon
To review a proposal, click the Edit icon 

PROPOSAL INFORMATION

TECHNICAL REVIEWS

☐ Edit proposal

Experiment

Proposal ID

Title

Abstract

Principal Investigator

Co-Proposers

Please give a brief summary of the proposal objective

Who have you contacted at the CLF to discuss your proposal?

Facility & Funding

Facility / Area

Number of weeks requested

Do you have any dates when you would be unavailable to carry out your experiment?

26110003

Laser Driven Ion Acceleration for Compact Medical Proton Sources

This experiment aims to exploit these capabilities to produce and characterise high energy proton beams from thin solid targets irradiated at intensities exceeding 10 10 W/cm , leveraging Astra Gemini s 30 fs, 15 J pulses and advanced diagnostics developed through past CLF campaigns . The primary objective is to map energy spectra, divergence, and stability of laser accelerated proton beams across a range of target materials and thicknesses, supported by multi angle ion spectrometry, time of flight detectors, and ultrafast optical probing. A secondary objective is to evaluate the feasibility of compact beam transport concepts relevant to applications such as medical proton therapy. By integrating knowledge from previous CLF experiments in particle acceleration, ultrafast plasma dynamics, and high contrast laser target interaction, this project aims to deliver new insights into optimising proton yield and beam quality. The outcomes will advance understanding of ion acceleration physics while supporting future development of compact, laser driven accelerator technologies.

Liam O'Connor (liamoconnor@utoronto.ca) (University of Toronto) (CANADA)

Thabo Mbeki (thabo.mbeki@manchester.ac.uk) (University of Manchester) (UNITED KINGDOM) Ana Silva (ana.silva@manchester.ac.uk) (University of Manchester)

1. Generate high energy proton beams using ultra short, high intensity laser pulses interacting with thin solid targets. 2. Characterize proton energy spectra using standard diagnostics such as RCF stacks and Thomson parabola spectrometers. 3. Investigate scaling laws for proton acceleration across different target materials and thicknesses. 4. Assess suitability for compact accelerator injection, including beam divergence, reproducibility, and flux stability.

Jobe Stevens

Gemini TA3

1

No

From the PROPOSAL INFORMATION tab, you can view all the Proposal's details.

To add your comments to the review, click TECHNICAL REVIEWS

You can DOWNLOAD A PDF copy of the proposal by clicking on the download button.

The proposal falls under the following science programmes	Engineering
Does this proposal have any industrial involvement/sponsorship?	No
Does your group have wider industry links not directly involving your present use of the Central Laser Facility?	No
Do you have any research grants associated with this proposal?	No
Publications	
Publications	New Scientist December 2023
Requirements	
Upload Technical Specification	Laser Driven Ion Acceleration for Compact Medical Proton Sources.pdf
Additional requirements	
Safety	
Additional Safety Considerations	Read the document
Science Case Upload	
Upload document	Laser Driven Ion Acceleration for Compact Medical Proton Sources Proposal.pdf
I confirm the above	Yes
Final Submission	
I have read and understand the Terms of Access	Yes

DOWNLOAD PDF

Internal reviews

Expand to see the internal reviews and reviewers



Assign to someone else?

If you think there is a better candidate to do the review for the proposal, you can re-assign it to someone else

Technical reviewer

Mei Ling



ASSIGN

Reviewed by Mei Ling

1

New technical review

NOTE: The sections **Status* and **Time allocation*** are for the FAP Secretary to complete**

Status *



Time allocation(Weeks) *



Internal comment

File Edit View Insert Format Tools

B

I

p

0 words

Internal documents

Accepted formats: .pdf

Maximum 5 file(s)

Comments for the review panel

File Edit View Insert Format Tools

B *I*

Comments for review panel are completed by the FAP Secretary.

p

0 words ↗

The remainder of the Technical Review is comprised of 8 sections included to capture comments providing the panel with pertinent information regarding how the experiment might be undertaken at HPL.

Please fill in any box that is relevant to you, there are boxes for Experimental Requests, Specialist Equipment, Hazards ect.

NOTE – Each section has its own text box in which to add the specific comments, (not shown in this documentation)

- Non-standard experimental requests (eg laser parameters, target fabrication, beam layouts, diagnostics)
- Long lead / expensive equipment (eg optics, large engineering)
- Specialist equipment not supplied by CLF (eg gas capillaries, magnets)
- Unusual hazards (eg electrical, chemical)
- Developments that are required for the experiment (eg new diagnostics)
- Developments that would be beneficial for the experiment
- Level of pre set-up required
- Useful knowledge from past experiments

Each of the 8 sections will have its own individual text box, (as seen below), where you add your comments.

Please fill in details below



You have **full rich text** *formatting* options for the comments _{you} ^{add}.

p

0 words

How much Engineering Effort does this require? *

Final Pdf upload
Accepted formats: .pdf

ATTACH FILE

All technical reviews will be exported to create a final pdf which will be uploaded here.

THIS WILL BE DONE BY THE FAP SECRETARY ONLY.

OK

RESET

SAVE

SAVE AND CONTINUE

Saving your comments on the Technical Review

× Fap - Proposal View

Please fill in details below

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Incorporating lessons from previous CLF campaigns allows us to:

- Refine laser and target parameters based on known acceleration performance and plasma behaviour.
- Justify diagnostic choices by referencing successful multi diagnostic setups from past studies.
- Improve experimental robustness using proven methods for alignment, contrast control, and shot sequencing.
- Enhance safety and engineering planning drawing from decades of experience handling extreme conditions.
- Demonstrate feasibility and readiness because similar physical regimes have already been accessed at CLF.

Once you have added your comments to the relevant section(s) and uploaded any important .pdf documentation you need to save the review for the next person, or if all the reviews have been completed, ready for the FAP review process.

74 words ↗

How much Engineering Effort does this require? *

1

Final Pdf upload
Accepted formats: .pdf



Safety Information.pdf

16.16 KB

 

+

ATTACH FILE

BACK

RESET

SAVE AND CONTINUE

To save you comments, click

SAVE AND CONTINUE

The saved Technical Review

✕ Fap - Proposal View

✔ Saved



PROPOSAL INFORMATION

TECHNICAL REVIEWS

A pop-up confirms your comments have been successfully saved.

Internal reviews

Expand to see the internal reviews and reviewers

Assign to someone else?

If you think there is a better candidate to do the review for the proposal, you can re-assign it to someone else

Technical reviewer

Mei Ling

ASSIGN

Reviewed by Mei Ling

Technical Review



New technical review

2

Review

New technical review

Technical Review ID	5263 (Pre-submission)
Status	FEASIBLE
Time allocation(Weeks)	1
Public Comment	Scientific quality and impact:The proposal addresses an active and relevant area of high-power laser science: laser-driven proton acceleration. The scientific goals are well defined, and the experiment is positioned to provide meaningful data on proton beam energy, divergence, and stability. The link to potential applications (e.g. compact accelerators) is appropriate without overstating near-term outcomes. Fit to facility:The experiment is well matched to CLF's flagship high-power laser systems. It demonstrates understanding of what the facility can realistically deliver and does not rely on speculative or unproven capabilities. The proposal makes effective use of CLF diagnostics and target areas. Deliverability and risk:The technical risk is moderate to low. The core experiment is viable without requiring major new developments or facility modifications, and optional enhancements are clearly separated from baseline objectives. The pre-set-up and safety considerations appear manageable within standard CLF processes. User preparedness:The team demonstrates familiarity with high-power laser experiments and appropriate planning

	team demonstrates familiarity with high-power laser experiments and appropriate planning for data acquisition, shot rate, and target usage. The proposal shows evidence of learning from prior work and has credible mitigation strategies for expected challenges such as shot-to-shot variability. Value for beamtime: The proposed work offers good scientific return for the requested beamtime, with the likelihood of producing publishable results and informing future experimental campaigns. The experiment also has potential to benefit the wider CLF user community through transferable methods and diagnostics.
You will see where the comments have been added to the Technical Review Internal Comment	Overall assessment: This is a well-conceived and technically sound proposal that is well aligned with existing CLF high-power laser capabilities, particularly Vulcan and Astra Gemini. The experiment is feasible within the requested beamtime and makes effective use of mature diagnostic techniques while offering scope for incremental development. Strengths: The proposed ion-acceleration study builds directly on a substantial body of previous CLF work in laser-plasma interaction and particle acceleration, demonstrating clear awareness of past experiments and known operational constraints. The experimental design is realistic, with a good balance between ambition and deliverability. The baseline experiment can proceed using standard CLF infrastructure, with optional enhancements clearly identified rather than assumed. Diagnostics are well chosen and appropriate for the scientific objectives, and the proposal shows a good understanding of shot-to-shot variability and the need for statistical data. Safety, engineering, and pre-set-up requirements are properly considered, with no obvious unmitigated risks. Technical feasibility: The requested laser parameters, target types, and diagnostics are within demonstrated operational regimes at CLF. No critical long-lead developments are required for the core experiment, reducing delivery risk. Any specialist equipment (e.g. transport magnets or structured targets) is appropriately identified as user-supplied. CLF benefit: The experiment reinforces CLF's role in ion acceleration research and has potential to generate reusable diagnostic configurations and experimental workflows for future users. Data produced will be valuable beyond the immediate application case. Recommendation: Strongly recommended for beamtime allocation. The experiment is technically credible, scientifically worthwhile, and suitable for execution within the proposed timeframe.
Internal Documents	
Please fill in details below	This experiment is likely to require at least the following non-standard items: Nanostructured or ...
Please fill in details below	For the proposed laser driven proton acceleration experiment, the only items that are potentially lo...
Please fill in details below	The following items WILL count as specialist, not supplied equipment: Required if using advanced bea...
Please fill in details below	The most probable unusual hazards you may need to declare are: Electrical: High voltage supplies fo...
Please fill in details below	Based on the current experimental concept, the most realistic development requirements are: Enhance...
Please fill in details below	For this specific experiment, the following developments would offer the best return on investment: ...
Please fill in details below	The proposed experiment involves laser driven proton acceleration using thin solid targets, with opt...

Please fill in details below

Incorporating lessons from previous CLF campaigns allows us to: Refine laser and target parameter...

How much Engineering Effort does this require?

1

Final Pdf upload

☐ Submitted

BACK

 SAVE

If you need to make amendments to the comments you can revert to an editable version of the review by clicking..

BACK

If you have any issues or questions about the process, please contact fase-support@stfc.ac.uk

Help and Support

For technical issues please contact, FASE-support@stfc.ac.uk

For questions regarding the review process please contact the FAP Secretary

The Central Laser Facility (CLF) at the Rutherford Appleton Laboratory is one of the world's leading laser facilities providing scientists from the UK and Europe with an unparalleled range of state of the art technology to carry out a broad range of experiments in physics, chemistry and biology. Our laser facilities range from advanced, compact tuneable lasers which can pinpoint individual particles to high power laser installations that recreate the conditions inside stars.

